GymX — Project Documentation (Phases 1–3)

Project Planning & Management · Literature Review · Requirements Gathering
Team 1 · MERN Stack · 8 Weeks

GymX — Stronger Everyday. A customizable MERN web platform that connects a gym with its members: memberships, QR attendance, equipment guides, gamified rewards and a private community, delivered by a 6-person team in 8 weeks.

Project at a Glance

Item	Detail
Project name	GymX — Smart Gym Experience
Team	Team 1 (6 members)
Team leader	Omar Sherif Bassiouny Abdelsamad
Track	Full-Stack Web Development
Technology stack	MERN — MongoDB, Express.js, React, Node.js
Duration	8 weeks
Repository	github.com/Omar-Sherif3444/GymX
Documentation scope	Phase 1 Project Planning & Management, Phase 2 Literature Review, Phase 3 Requirements Gathering

Team Members

#	Student name	Student ID	Email	Primary role
1	Omar Sherif Bassiouny Abdelsamad	21148715	omarsherif3444@gmail.com	Team Leader, Project Manager, Backend Lead
2	Ahmed Mostafa Khalil Abdelwahed	21154697	ahmed.mostafa.2242007@gmail.com	Backend Developer (Auth, Memberships, Attendance)
3	Waad Ali Abdelghany Hafez	21164012	waada7710@gmail.com	UI/UX Designer, Frontend Lead
4	Amr Ahmed Sayed Ibrahim	21171922	amrahmed201135@gmail.com	Backend Developer (Database, Gamification, Community)
5	Farida Mohamed Ahmed Shawky	21183159	faridashawky007@gmail.com	Frontend Developer (Member & Admin Dashboards)
6	Khaled Ahmed Hamdy Abdelkhalek	21169283	khaled882005@gmail.com	QA Engineer, DevOps, Documentation

Every member works across the stack on the features they own; the primary role says who leads each area.

Brand Identity

The visual theme is taken from the approved GymX logo: an X formed by four crossed dumbbells around a central diamond, with the wordmark "GYM X" and the tagline "Stronger Everyday".

Token	Hex	Use in the website
Orange (primary)	#FF5A1F	Buttons, highlights, active states, the "X" in the wordmark, progress and reward accents
Dark	#0F0F0F	Dark-mode background, primary text in light mode, navigation bar
Light	#F6F4EE	Light-mode background, primary text in dark mode, cards

The site supports both a dark and a light theme, matching the two logo variants. Typography uses a wide geometric sans-serif for headings (matching the wordmark) and a clean sans-serif for body text. The square dumbbell mark serves as the app icon and favicon.

1. Project Planning & Management

GymX will deliver a working, deployed MERN platform in 8 weeks using Scrum: 2 weeks of planning and design, two 2-week development sprints, then 1 week of testing and 1 week of release.

1.1 Project Idea and Problem Statement

Based on the team members' own experience as gym members in Egypt, many small and mid-sized local gyms still appear to run on paper membership cards, Excel sheets, WhatsApp groups and Facebook pages. Members cannot easily check when their subscription ends or how often they have attended. Most members, especially beginners, are not sure how to use each machine

safely or which muscles it trains, and many lose motivation after the first few weeks. These observations have not yet been measured; the planned interviews and questionnaire (Section 3.1) will test them.

GymX solves this with one customizable website per gym. Members get a personal account with their subscription, a QR code, their attendance history, a guide to the gym's own equipment, rewards for staying active and a private community. Gym management gets one place to publish its equipment and services, record attendance and keep members engaged.

1.2 Project Objectives

1. Deliver a secure member account where each member sees subscription status, expiry date, remaining days and a personal QR code.
2. Replace paper attendance with QR-based check-in and check-out recorded by gym staff, with a full history for each member.
3. Publish an equipment guide per gym that explains the targeted muscles, correct usage steps and example exercises for each machine.
4. Increase member consistency through tokens, challenges, badges and a leaderboard.
5. Build a private, moderated community feed where members of the same gym share achievements and join challenges.
6. Let each gym customize the platform: name, logo, equipment list, services, working hours and challenges.
7. Deploy the system online and present it within 8 weeks, with complete documentation.

1.3 Project Scope

In scope (version 1.0)

- User registration, login and role-based access for Members, Gym Staff, Gym Admins and a Platform Super Admin.
- Gym profile and customization: branding, equipment, services, working hours.
- Membership plans and member subscriptions (start date, end date, status, remaining days, renewal history). Payments are recorded manually by staff.
- Personal member QR code (a signed, short-lived token validated on the server) and staff check-in / check-out by scanning it from the admin panel, with manual search as a fallback.
- Attendance history for members and attendance reports for admins.
- Equipment guides with muscle groups, instructions, safety tips, images or video links, and exercises.
- Gamification: tokens for attendance and challenges (Must); badges and a gym leaderboard (Should); side quests, a mini-game and token redemption (Could, only if time allows).
- Private community feed per gym: posts, images, likes, comments, achievement sharing and reporting of inappropriate posts.
- Admin dashboard with key numbers: active members, expiring subscriptions, daily check-ins.
- Responsive web interface with dark and light themes based on the GymX brand.

Out of scope (version 1.0)

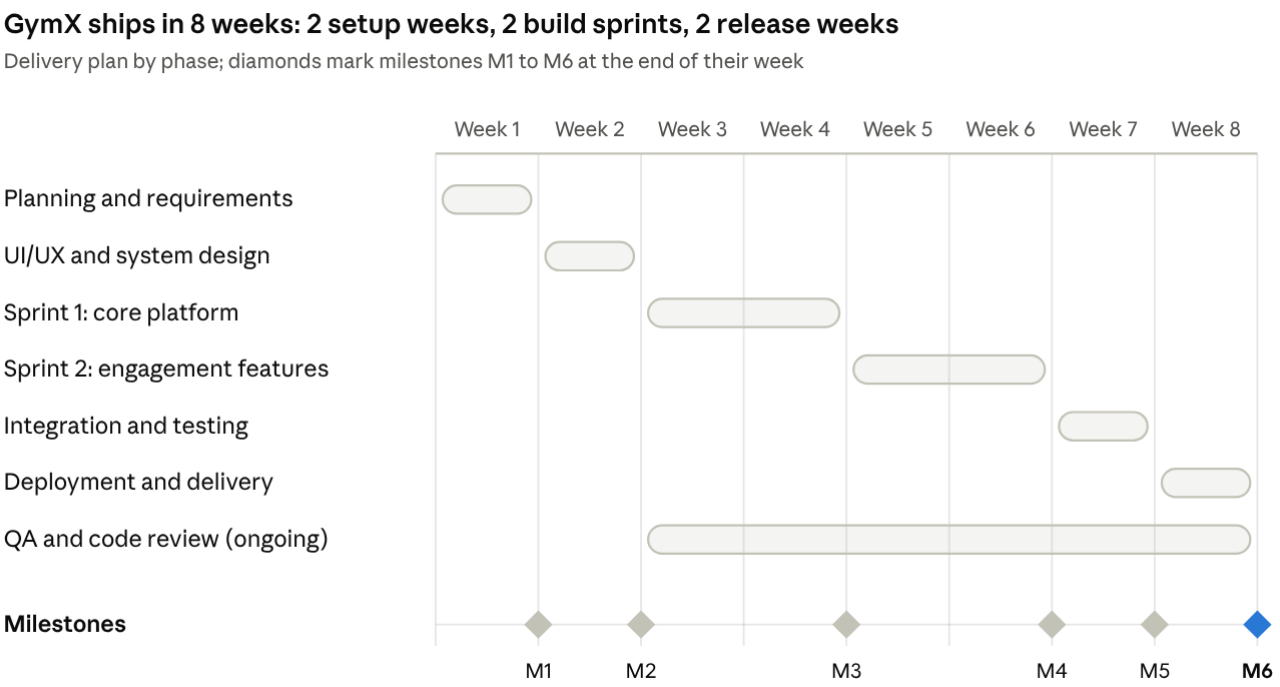
- Online payment gateways (e.g. Paymob, Fawry, Stripe).
- Native iOS / Android apps (the website is mobile-responsive instead).
- Turnstile, kiosk or door hardware integration.
- Wearable and health-app integration (Apple Health, Google Fit, smartwatches).
- AI-generated workout or nutrition plans.
- Redeeming tokens for real products or discounts with external partners.

1.4 Methodology and Tools

The team follows Agile Scrum with two 2-week sprints, a short daily stand-up (online, 15 minutes), a sprint review and a retrospective at the end of each sprint. Omar Sherif acts as Scrum Master and Product Owner representative.

Purpose	Tool
Source control and code review	Git, GitHub (branches `main`, `develop`, `feature/*`; every pull request needs 1 approving review)
Task board	GitHub Projects (Kanban: Backlog, To Do, In Progress, Review, Done)
UI/UX design	Figma
API design and testing	Postman, Swagger / OpenAPI
Database	MongoDB Atlas, Mongoose
Communication	WhatsApp group for quick updates, Google Meet / Discord for meetings
Deployment	Vercel or Netlify (React frontend), Render or Railway (Node / Express API), MongoDB Atlas (database)
Documentation	This document, README in the repository

1.5 Project Timeline (8 Weeks)



Sprints 1 and 2 carry the build in weeks 3 to 6; QA and code review run continuously from week 3 until final delivery at M6.

Milestone	End of week	Exit criteria
M1 Requirements approved	1	Phases 1 to 3 of the documentation submitted and approved
M2 Design freeze	2	Figma screens, database schema and API endpoint list agreed; repository and project board set up
M3 Core platform ready	4	Auth, roles, gym profile, memberships, QR check-in and attendance working end to end
M4 Feature complete	6	Equipment guides, gamification, community and admin dashboard working end to end
M5 Release candidate	7	All critical and major bugs fixed, tests passing, security checklist done
M6 Final delivery	8	System deployed online, final documentation and demo presentation delivered

Weekly work plan

Week	Focus	Key deliverables
1	Planning and requirements	Project plan, literature review, requirements, user stories, product backlog
2	UI/UX and system design	Wireframes and high-fidelity Figma screens, ERD / MongoDB schemas, REST API design, repo and CI setup
3	Sprint 1, part 1	Auth with JWT, roles, gym profile and customization, base React layout and theme
4	Sprint 1, part 2	Membership plans and subscriptions, QR generation, check-in / check-out, attendance history
5	Sprint 2, part 1	Equipment guides (admin CRUD and member pages), token engine, challenges
6	Sprint 2, part 2	Community feed, admin dashboard, badges and leaderboard; side quest only if all Must-haves are done
7	Integration and testing	End-to-end testing, bug fixing, performance and security hardening, user acceptance test with real gym members
8	Deployment and delivery	Production deployment, final documentation, demo video and presentation

Scope control. The 29 Must-have requirements (Section 3.4) form the minimum product and are split across the two sprints: 14 in Sprint 1 (accounts, gym setup, memberships, QR attendance) and 15 in Sprint 2 (equipment guides, tokens and challenges, community, admin dashboard). Should-have items are built only after a sprint's Must-haves pass review. Could-have items (side quests and mini-game, token redemption, email notifications, subscription freeze, pinned announcements, peak-time charts, starter equipment library) are dropped first if the schedule slips.

1.6 Task Distribution and Responsibilities

Member	Main responsibilities	Sprint 1 ownership	Sprint 2 ownership
Omar Sherif (Leader)	Planning, Scrum, architecture, code reviews, Git workflow, integration	Project structure, Express setup, JWT auth and role middleware	Admin dashboard API, integration of all modules
Ahmed Mostafa	Backend development	Membership plans and subscriptions API, QR generation	Equipment guides API, image upload
Waad Ali	UI/UX design, frontend lead	Design system (colors, typography, components), auth and landing pages	Gamification pages: challenges, badges, leaderboard

Member	Main responsibilities	Sprint 1 ownership	Sprint 2 ownership
Amr Ahmed	Database and backend development	MongoDB schemas, check-in / check-out and attendance API	Token engine, challenges and community feed API
Farida Mohamed	Frontend development	Member dashboard: subscription, QR code, attendance history	Equipment guide pages, community feed UI, admin dashboard UI
Khaled Ahmed	QA, DevOps, documentation	Test plan, Postman collections, CI with GitHub Actions	End-to-end tests, deployment, final documentation

Each feature has one owner, and every pull request is reviewed by a second member before merging into `develop`.

1.7 Risk Management

Risk	Likelihood	Impact	Mitigation / response
Scope creep: too many features for 8 weeks	High	High	Freeze the version 1.0 scope at M2; extra ideas go to the "Future" backlog; prioritize with MoSCoW
Members' schedules clash with exams and other courses	High	Medium	Plan around exam weeks, keep tasks small (1 to 3 days), pair a backup owner for every feature
Uneven MERN experience in the team	Medium	Medium	Week 2 knowledge-sharing sessions, shared coding standards, pair programming on hard tasks
Integration problems between frontend and backend	Medium	High	Agree the API contract in week 2 (Swagger), use mock data early, integrate continuously instead of at the end
Merge conflicts and lost work in Git	Medium	Medium	Feature branches, small pull requests, protected `main` branch, daily pulls from `develop`
Security weaknesses (stolen accounts, fake check-ins)	Medium	High	Hashed passwords, JWT expiry, role checks on every route, signed short-lived QR tokens, server-side membership validation, duplicate check-in blocking, input validation, rate limiting
A team member leaves or becomes unavailable	Low	High	Shared documentation, code reviews so at least 2 people know each module, redistribute tasks at sprint planning
Free hosting limits or deployment failure	Medium	Medium	Deploy a staging version from week 4, keep environment variables documented, have a second hosting option ready
Low adoption of the community or gamification features	Medium	Medium	Test with real gym members in week 7, keep interactions simple, seed sample challenges and content

1.8 Success Metrics (KPIs)

KPI	Target	How it is measured
Must-have requirements delivered	100% of Must-have, at least 70% of Should-have	Requirements checklist at M6
Schedule adherence	All 6 milestones met within 3 days of their planned week	Milestone table and GitHub Projects board
Sprint completion rate	At least 80% of committed sprint tasks done	Sprint review
Backend test coverage	At least 70% of API services and controllers	Jest coverage report
Critical bugs at release	0 open critical bugs, at most 5 minor bugs	GitHub Issues
Performance	Target: pages load in under 3 seconds; 95% of API responses under 500 ms with 100 concurrent users	Lighthouse and Postman / load-test results
Usability	Average SUS score of at least 70 from at least 10 test users	System Usability Scale questionnaire in week 7
Check-in speed	A staff member completes a QR check-in in under 5 seconds	Timed test at reception
Team collaboration	Every member has merged pull requests in both sprints	GitHub contribution history

All KPI values are targets set at the planning stage, not guaranteed results. Actual values will be measured during week 7 testing and reported at M5 and M6.

2. Literature Review

Among the 9 solutions reviewed, none appeared to combine membership and attendance tracking with a guide to the gym's own equipment, gym-scoped gamification and a private community in the single gym-specific workflow GymX proposes. GymX is designed to fill that gap, within the limits of a review based on public information (Section 2.1).

2.1 Review Approach

The team reviewed 9 existing solutions in three groups, plus published research on gamification:

1. **Gym management platforms** used by gym owners: Mindbody, ABC Glofox, Virtuagym, PushPress.
2. **Member-side fitness apps** used by individuals: Hevy, Strava, MuscleWiki.
3. **Local and academic solutions**: the Egyptian "Gym Manager" app and a gym management graduation project from An-Najah National University.

Each solution was assessed against the same 8 criteria used in Section 2.4: (1) membership and subscription management, (2) attendance or check-in, (3) exercise or equipment guidance, (4) equipment specific to the member's gym, (5) gamification, (6) a community private to one gym, (7) per-gym branding, and (8) cost for a small gym.

Sources and limitations. The review relies on vendor websites, app-store listings, third-party review sites and one peer-reviewed study, all accessed in October 2026 [1]–[18]. The team did not run hands-on trials of the commercial products, so ratings reflect public descriptions and may miss unlisted or recently added features. Prices are in US dollars for US plans, change often and may differ in Egypt. Statements marked "team assessment" are the authors' interpretation, not a sourced fact. The sample of 9 solutions is not exhaustive and under-represents tools used in Egypt.

2.2 Existing Solutions: Strengths and Weaknesses

Solution	Type	Strengths	Weaknesses
Mindbody	Gym / studio management (USA)	Scheduling, memberships, POS, reporting and staff management; a consumer marketplace that exposes the business to new clients; branded app as an optional add-on [2]	US plans start at $79/month per location, with higher tiers quote-based; reviewers note a steep learning curve and contract terms [2]; pricing is a frequent user complaint [3]
ABC Glofox	Gym / studio management	Scheduling, memberships, billing, access control and check-in, CRM and reporting; custom branded member app; member community tools; used by 30,000+ studios and gyms [1]	Positioned mainly for class-based studios; prices not published; no equipment guide or gamification listed among its features [1]
Virtuagym	Gym management + coaching	4,000+ exercises, workout templates and gamification for goals and achievements [4]; nutrition tools and branded member apps [5]	Setup and configuration heavier than simpler tools; no free version [5]; generic exercise library rather than the gym's own machines (team assessment)
PushPress	Gym management	Self check-in by tapping a name; member portal; check-in and revenue reports [6]; free plan for up to 100 members, Pro $159/month, Max $229/month [7]	Aimed mainly at class-based gyms such as CrossFit affiliates and bootcamps [6]; no equipment education among its listed features [7]
Hevy	Member workout tracker	Social and discovery feeds, routine sharing, performance comparison [8]; free tier with unlimited logging, full exercise library and social feed [9]	Not linked to a gym's memberships, attendance or equipment (team assessment); the discovery feed shows workouts from users outside the member's gym [8]
Strava	Social fitness network	Clubs, monthly challenges, badges and leaderboards [10]	Built around running and cycling [10]; group challenges were limited to 3 for free users at the time of reporting [11]
MuscleWiki	Exercise library (web + app)	Free directory browsable by muscle and equipment, with video demonstrations and step-by-step form guidance [12]; 2,000+ exercises [18]	Generic: not linked to the machines a specific gym owns; no membership, attendance or community features (team assessment)
Gym Manager (مدير الجيم)	Local gym management app (Egypt)	Arabic and English; renewal dates and renewal; communication with coaches and members; session booking to reduce crowding [13]	iPhone only; too few ratings to show adoption [13]; no equipment guides, gamification or community feed in its description [13]
Gym Management App (An-Najah National University)	Academic graduation project	Web and mobile; member registration, workout and nutrition plans, billing, supplements store, reports and analytics [14]	Management-focused; no gamification, private community or per-gym equipment guide described in its abstract [14]

2.3 Research on Gamification and Community

Published research supports GymX's engagement features. A meta-analysis of 16 randomized controlled trials with 2,407 participants found a small-to-medium positive effect of gamified interventions on physical activity (Hedges g = 0.42), and the effect persisted after follow-up, suggesting it was not only a novelty effect [15]. Platform data from Strava points the same way for communities: in 2016, users who joined a club within 14 days of signing up uploaded 116.3% more activities over the next six months than the average user [10]. That figure is company data reported by a magazine, not a controlled study, so it shows an association rather than a cause.

2.4 Feature Comparison with GymX

Solution	Membership & subscription	Attendance / check-in	Exercise or equipment guide	Gym-specific equipment	Gamification	Private gym community	Per-gym branding	Free / low cost for small gyms
Mindbody [2]	Yes	Yes	No	No	No	No	Partial (add-on)	No
ABC Glofox [1]	Yes	Yes	No	No	No	Yes	Yes	No
Virtuagym [4], [5]	Yes	Yes	Yes	Partial	Yes	Partial	Yes	No
PushPress [6], [7]	Yes	Yes	No	No	Partial	Partial	Partial	Partial (free plan)
Hevy [8], [9]	No	No	Yes	No	Partial	No (public)	No	Yes

Solution	Membership & subscription	Attendance / check-in	Exercise or equipment guide	Gym-specific equipment	Gamification	Private gym community	Per-gym branding	Free / low cost for small gyms
Strava [10], [11]	No	No	No	No	Yes	Partial (clubs)	No	Partial
MuscleWiki [12]	No	No	Yes	No	No	No	No	Yes
Gym Manager (Egypt) [13]	Yes	Partial	No	No	No	No	No	Yes
GymX (planned)	**Planned**	**Planned (QR)**	**Planned**	**Planned**	**Planned**	**Planned**	**Planned**	**Planned**

Ratings are the team's reading of the cited sources in October 2026, not results of hands-on testing. "Partial" means the feature exists in a limited form or only on some plans; "No" means the feature was not found in the sources reviewed. GymX values are planned features, not delivered results.

2.5 Gap Analysis and Lessons Learned

Based on the sources reviewed, management platforms (Mindbody, Glofox, PushPress) cover the business side well but appear to offer little guidance to members inside the gym [1], [2], [6]. Member apps (Hevy, Strava, MuscleWiki) motivate and educate but are not linked to the member's gym, subscription or equipment [8], [10], [12]. The local and academic solutions reviewed focus on management [13], [14]. GymX aims to connect both sides for one gym at a time.

Lesson from	What GymX adopts	What GymX avoids
ABC Glofox	Per-gym branding (logo, colors) so each gym's members see their own gym	Building separate native apps per gym; GymX uses one responsive web app with themes
PushPress	Check-in must take seconds; GymX targets a QR scan under 5 seconds	Self check-in by name only, which is easy to misuse; GymX uses signed, short-lived QR tokens validated on the server
Mindbody	A clear admin dashboard with key numbers	Complex admin screens and high cost for small gyms
Virtuagym, MuscleWiki	Browse exercises by muscle group with step-by-step instructions and video links	Thousands of generic exercises; GymX shows only the machines the gym actually owns
Strava	Gym-scoped challenges, badges and a leaderboard	Pay-walled challenges; core challenges stay free for members
Hevy	Achievement sharing and a simple feed with likes and comments	A public feed; GymX keeps the community private to verified members of the same gym
Gym Manager (Egypt)	Expiry and renewal visibility for members; local market focus	iPhone-only delivery; GymX works in any browser on any device
Mazeas et al. (2022)	Use proven game elements: goals, rewards, challenges, feedback	Gamification without real goals; every token is tied to attendance or a fitness activity

3. Requirements Gathering

GymX version 1.0 defines 5 user roles, 51 functional requirements across 9 modules (29 Must-have) and 22 quality requirements with measurable targets. These requirements come from team brainstorming and the competitor analysis; they remain provisional until the planned interviews and questionnaire are conducted.

3.1 How Requirements Were Gathered

#	Technique	Status
1	**Brainstorming** within the team, based on the project idea and the members' own gym experience	Completed
2	**Competitor analysis** of 9 existing solutions (Section 2)	Completed
3	**Interviews** with gym owners or managers and reception staff at local gyms, to confirm the admin and check-in workflows	Planned (week 1), not yet conducted
4	**Questionnaire** for gym members (Google Forms), to rank the member features	Planned (week 1), not yet conducted
5	**Prioritization** of every requirement with MoSCoW: Must, Should, Could, Won't (this version)	Completed, to be reviewed after items 3 and 4

The results of the interviews and questionnaire, including the number of participants and any changes they cause to the requirements, will be documented in an appendix once they have been carried out.

3.2 Stakeholders and Their Needs

Stakeholder	Description	Main needs
Gym member (beginner)	New to training, unsure how machines work	Clear equipment instructions, safe technique, motivation to keep coming
Gym member (regular)	Trains several times a week	Quick check-in, subscription and attendance status, challenges, leaderboard, sharing achievements
Gym staff (reception / coach)	Handles daily operations at the front desk and gym floor	Fast QR check-in and check-out, find a member quickly, see if a subscription is active

Stakeholder	Description	Main needs
Gym admin (owner / manager)	Runs the gym and its platform	Customize the gym's page, manage members, plans, equipment and challenges, see attendance and expiry reports, moderate the community
Platform super admin (GymX team)	Operates the GymX platform	Onboard new gyms, create gym admin accounts, monitor the system
Project supervisors / evaluators	Assess the graduation / track project	A complete, documented, working and deployed system

3.3 User Roles and Permissions

Capability	Member	Staff	Gym admin	Super admin
View own subscription, QR code, attendance	Yes	No	No	No
Browse equipment guides	Yes	Yes	Yes	No
Join challenges, earn tokens, post in community	Yes	No	No	No
Scan QR for check-in / check-out	No	Yes	Yes	No
Register members and record subscriptions	No	Yes	Yes	No
Manage plans, equipment, challenges, gym branding	No	No	Yes	No
Moderate community posts	No	No	Yes	No
View gym reports and dashboard	No	Limited	Yes	No
Create gyms and gym admin accounts	No	No	No	Yes

A visitor (not logged in) can see a gym's public landing page: services, equipment overview, working hours and contact details.

3.4 Functional Requirements

FR-1 Authentication and Accounts

ID	Requirement	Priority
FR-1.1	The system shall let members register with name, email, phone and password, linked to one gym (by gym invitation code or added by staff).	Must
FR-1.2	The system shall let all users log in and log out securely.	Must
FR-1.3	The system shall restrict every page and API endpoint by role (member, staff, gym admin, super admin).	Must
FR-1.4	The system shall let users reset a forgotten password by email.	Should
FR-1.5	The system shall let users edit their profile (photo, phone, fitness goal).	Should

FR-2 Gym Customization

ID	Requirement	Priority
FR-2.1	The super admin shall create a gym and its first gym admin account.	Must
FR-2.2	The gym admin shall set the gym's name, logo, accent color, address, working hours and contact details.	Must
FR-2.3	The gym admin shall publish the list of services and facilities (e.g. cardio area, sauna, classes).	Should
FR-2.4	The system shall show a public landing page for each gym using its branding.	Should

FR-3 Memberships and Subscriptions

ID	Requirement	Priority
FR-3.1	The gym admin shall create, edit and deactivate membership plans (name, duration, price, description).	Must
FR-3.2	Staff shall assign a plan to a member with a start date; the system shall calculate the end date.	Must
FR-3.3	Members shall see their plan, start date, end date, status (active, expiring soon, expired) and remaining days.	Must
FR-3.4	Staff shall renew a subscription and the system shall keep the full subscription history.	Must
FR-3.5	Staff shall freeze a subscription for a set number of days, extending its end date.	Could
FR-3.6	The system shall notify members 7 days and 1 day before their subscription expires.	Should

FR-4 QR Code and Attendance

ID	Requirement	Priority
FR-4.1	The system shall show each member a QR code containing a signed token (no raw IDs). The token expires 2 minutes after it is generated and refreshes automatically every 60 seconds while the QR page is open.	Must
FR-4.2	Staff shall scan a member's QR code with the device camera to record check-in, and scan again to record check-out.	Must
FR-4.3	On every scan, the server shall verify the token's signature and expiry, check that the member belongs to the scanning staff member's gym, and confirm the subscription is active and not frozen; otherwise it shall reject the check-in and show the reason to staff.	Must
FR-4.4	Staff shall check members in manually by name or phone when a QR code cannot be scanned; manual check-ins are logged with the staff member's name.	Should
FR-4.5	Members shall see their attendance history with date, check-in time, check-out time and duration.	Must
FR-4.6	The system shall show members their visit statistics (visits this month, current weekly streak).	Should

ID	Requirement	Priority
FR-4.7	The gym admin shall view and export attendance reports by day, week or month.	Should
FR-4.8	The system shall prevent duplicate check-ins: each token can be used only once, a member already checked in cannot be checked in again until checked out, and visit tokens are awarded at most once per visit.	Must
FR-4.9	The gym admin shall be able to reset a member's QR secret, invalidating all earlier codes, if misuse is suspected.	Should

FR-5 Equipment Guides

ID	Requirement	Priority
FR-5.1	The gym admin shall add, edit and remove the equipment available in the gym, with name, photo and category.	Must
FR-5.2	Each equipment page shall show the targeted primary and secondary muscles.	Must
FR-5.3	Each equipment page shall show step-by-step usage instructions and safety tips.	Must
FR-5.4	Each equipment page shall list exercises and techniques, with optional video links.	Should
FR-5.5	Members shall search and filter equipment by name, category or muscle group.	Must
FR-5.6	The system shall provide a starter library of common equipment that gyms can copy and edit.	Could

FR-6 Gamification and Rewards

ID	Requirement	Priority
FR-6.1	The system shall award tokens automatically for each valid gym visit.	Must
FR-6.2	The gym admin shall create challenges with a goal (e.g. 12 visits in 30 days), dates and a token reward.	Must
FR-6.3	Members shall join challenges and track their progress.	Must
FR-6.4	The system shall keep a token balance and a transaction history for every member.	Must
FR-6.5	The system shall award badges for milestones (e.g. first visit, 50 visits, 4-week streak).	Should
FR-6.6	The system shall show a monthly gym leaderboard by tokens.	Should
FR-6.7	The system shall offer side quests and one mini-game (e.g. a daily fitness quiz) that award tokens.	Could
FR-6.8	The gym admin shall define rewards that members can redeem with tokens (e.g. a free guest pass), confirmed by staff.	Could

FR-7 Private Gym Community

ID	Requirement	Priority
FR-7.1	Members shall see a community feed visible only to members of the same gym.	Must
FR-7.2	Members shall create posts with text and an optional image.	Must
FR-7.3	Members shall like and comment on posts.	Must
FR-7.4	The system shall let members share an earned badge or completed challenge to the feed in one click.	Should
FR-7.5	Members shall report inappropriate posts; the gym admin shall hide or delete posts.	Must
FR-7.6	The gym admin shall pin announcements at the top of the feed.	Could

FR-8 Admin Dashboard and Member Management

ID	Requirement	Priority
FR-8.1	The gym admin shall see a dashboard with active members, subscriptions expiring within 7 days and today's check-ins.	Must
FR-8.2	Staff and admins shall search, view, add and deactivate member accounts.	Must
FR-8.3	The gym admin shall add and remove staff accounts.	Must
FR-8.4	The dashboard shall show attendance trends by day and by hour (peak times).	Could

FR-9 Notifications

ID	Requirement	Priority
FR-9.1	The system shall show in-app notifications for expiry reminders, new challenges, badges earned, and likes or comments on a member's posts.	Should
FR-9.2	The system shall send email for password reset and subscription expiry reminders.	Could
FR-9.3	The system shall not send marketing messages; members control notification settings.	Should

Won't have in version 1.0: online payments, native mobile apps, hardware turnstile integration, wearable integration, AI workout plans (see Section 1.3).

3.5 Key User Stories

ID	User story	Acceptance criteria
US-1	As a member, I want to see how many days remain in my subscription so that I renew on time.	The dashboard shows end date, remaining days and an "expiring soon" label within 7 days of expiry.
US-2	As a member, I want to show my QR code at reception so that I enter quickly.	The QR code opens in 1 tap from the member dashboard and refreshes automatically every 60 seconds; a screenshot older than 2 minutes is rejected.

ID	User story	Acceptance criteria
US-3	As a receptionist, I want to scan a member's QR code so that check-in is recorded automatically.	Valid code: check-in saved and confirmed, with a target time under 5 seconds. Expired subscription, expired or reused code, or a member already checked in: check-in blocked with a clear message.
US-4	As a beginner, I want to see how to use a machine so that I train safely.	The equipment page shows muscles, numbered steps, safety tips and at least one exercise.
US-5	As a member, I want to join a challenge so that I stay motivated.	Progress updates after each visit; tokens are added automatically when the goal is reached.
US-6	As a member, I want to share my new badge with my gym community so that others can cheer me on.	One click creates a feed post visible only to members of the same gym.
US-7	As a gym admin, I want to add our machines so that members see only equipment we have.	Equipment added by the admin appears in the members' guide immediately.
US-8	As a gym admin, I want to see who expires this week so that staff can contact them.	The dashboard lists subscriptions ending within 7 days with member name and phone.

3.6 Non-Functional (Quality) Requirements

All numeric values below are design targets set before development. They will be measured during week 7 testing and reported with the actual results; they are not guaranteed outcomes.

ID	Category	Requirement	How it is verified
NFR-1	Performance	Main pages load in under 3 seconds on a 4G connection.	Lighthouse performance score of at least 80
NFR-2	Performance	Target: 95% of API requests respond in under 500 ms with 100 concurrent users.	Load test in week 7 (e.g. k6 or Artillery); result reported, not assumed
NFR-3	Performance	A QR check-in completes in under 5 seconds from scan to confirmation.	Timed test at reception
NFR-4	Security	Passwords are hashed with bcrypt (minimum 10 salt rounds); plain passwords are never stored or logged.	Code review
NFR-5	Security	Authentication uses JWT access tokens that expire within 1 day, stored in httpOnly cookies.	Code review, manual test
NFR-6	Security	Every protected endpoint checks the user's role and gym, so a user can never read another gym's data.	Automated authorization tests
NFR-7	Security	QR codes carry a signed token (not raw IDs) that expires 2 minutes after generation and rotates every 60 seconds, so forged codes are rejected and copied codes or screenshots stop working quickly. Each token is single-use, and membership status is always checked on the server, never trusted from the code.	Tests with tampered, expired and reused codes
NFR-8	Security	All input is validated on the server; the API is protected against the OWASP Top 10 risks (injection, XSS, broken access control).	Helmet, express-rate-limit, input validation, review against the OWASP Top 10
NFR-9	Security	Login is limited to 5 failed attempts per 15 minutes per account / IP.	Manual test
NFR-10	Security	All traffic uses HTTPS in production.	Deployment check
NFR-11	Privacy	Members' phone numbers, attendance and posts are visible only to authorized users of their own gym; members can delete their posts and request account deletion.	Authorization tests
NFR-12	Usability	A new member can find their subscription status and QR code within 2 clicks of logging in.	Usability test
NFR-13	Usability	Average System Usability Scale (SUS) score of at least 70.	SUS questionnaire with at least 10 users
NFR-14	Usability	The interface is responsive from 360 px phones to desktop screens and supports dark and light themes.	Testing on phone, tablet and desktop
NFR-15	Accessibility	Text contrast meets WCAG 2.1 AA (at least 4.5:1); images have alt text; all main actions work with a keyboard.	Lighthouse accessibility score of at least 90
NFR-16	Reliability	The system is available at least 99% of the time during the evaluation period.	Uptime monitor
NFR-17	Reliability	The database is backed up daily (MongoDB Atlas backups).	Atlas backup settings
NFR-18	Scalability	The design supports many gyms in one database (every record carries a gym ID), with indexes on frequent queries.	Schema review
NFR-19	Maintainability	Code follows a shared ESLint + Prettier style, a layered structure (routes, controllers, services, models) and documented REST endpoints (Swagger).	Pull-request reviews
NFR-20	Maintainability	At least 70% test coverage of backend services and controllers.	Jest coverage report
NFR-21	Compatibility	Works on the latest two versions of Chrome, Edge, Firefox and Safari, on Android and iOS.	Cross-browser testing
NFR-22	Localization	All interface text is kept in translation files so Arabic (RTL) can be added later.	Code review

3.7 Technical Requirements (MERN Stack)

Layer	Technology	Purpose
Frontend	React (Vite), React Router, Axios	Single-page application and API calls
Styling	CSS variables / Tailwind CSS with the GymX color tokens	Dark and light themes, responsive layout
QR	qrcode.react (generate), html5-qrcode (scan with camera)	Member QR codes and staff scanner
Backend	Node.js, Express.js	REST API
Database	MongoDB Atlas with Mongoose	Data storage and schemas
Auth and security	JWT, bcrypt, Helmet, express-rate-limit, Joi / Zod validation	Secure login and API protection
File storage	Multer + Cloudinary	Equipment photos, profile and post images
Testing	Jest, Supertest, React Testing Library, Postman	Unit, API and UI tests
CI / CD	GitHub Actions	Lint and test on every pull request
Hosting	Vercel / Netlify (frontend), Render / Railway (backend)	Production deployment

Main data entities: User (with role and gym), Gym, MembershipPlan, Subscription, Attendance, Equipment, Exercise, Challenge, ChallengeParticipation, TokenTransaction, Badge, Post, Comment, Report, Notification.

Constraints: 8-week schedule, a team of 6 students, free tiers of hosting and database services, and no online payment integration in version 1.0.

3.8 Design and Branding Requirements

- The interface shall use the GymX palette: Orange #FF5A1F for primary actions and highlights, Dark #0F0F0F and Light #F6F4EE for backgrounds and text.
- The site shall offer dark and light themes, matching the two official logo variants, and remember the user's choice.
- Headings shall use a wide geometric sans-serif matching the "GYM X" wordmark; body text shall use a readable sans-serif.
- The dumbbell X mark shall be used as the favicon and app icon; the full logo with "Stronger Everyday" appears on the landing and login pages.
- Each gym's own logo and accent color appear inside its member area, while GymX branding stays on the platform pages.
- Key screens to design in Figma: landing page, login / register, member dashboard, QR code, attendance history, equipment list and detail, challenges and leaderboard, community feed, staff scanner, admin dashboard, gym settings.

References

All online sources were accessed on 18 October 2026. Bracketed numbers in the text, such as [2], refer to this list.

1. ABC Glofox. *Features: fitness studio and gym management software*. https://www.glofox.com/features/
2. SchedulingKit. *Mindbody Review: Pricing, Pros & Cons (2026)*. https://schedulingkit.com/reviews/mindbody-review
3. Software Advice. *Mindbody Software Reviews, Demo & Pricing (2026)*. https://www.softwareadvice.com/club-management/mindbody-profile/
4. Techjockey. *Virtuagym Software Pricing, Features & Reviews*. https://www.techjockey.com/detail/virtuagym
5. FitGap. *Virtuagym*. https://us.fitgap.com/products/022775/virtuagym
6. PushPress. *Gym management software features*. https://www.pushpress.com/features
7. Software Advice. *PushPress Review: pricing and plans (2026)*. https://www.softwareadvice.com/membership-management/pushpress-profile
8. Hevy. *Hevy's Social Features*. https://www.hevyapp.com/features/social-features/
9. WhistleOut. *Hevy app review*. https://www.whistleout.com/CellPhones/Apps/hevy-app-review
10. Coach. *Eight Essential Tips for Strava Beginners*. https://www.coachweb.com/fitness-trackers/6314/eight-essential-tips-for-strava-beginners
11. Coach. *Strava's Group Challenges Lets You Compete Against Friends*. https://www.coachweb.com/fitness-apps/8942/strava-group-challenges
12. MuscleWiki. *Exercise library*. https://musclewiki.com/exercises
13. Apple App Store. *Gym Manager (مدير الجيم)*. https://apps.apple.com/us/app/id1555868079
14. An-Najah National University Repository. *Gym Management App (graduation project)*. https://repository.najah.edu/items/f8899626-2099-4386-8840-d837b2f30b9d/full
15. Mazeas, A., Duclos, M., Pereira, B., & Chalabaev, A. (2022). Evaluating the Effectiveness of Gamification on Physical Activity: Systematic Review and Meta-analysis of Randomized Controlled Trials. *Journal of Medical Internet Research*, *24*(1), e26779. https://doi.org/10.2196/26779
16. OWASP Foundation. *OWASP Top Ten*. https://owasp.org/www-project-top-ten/
17. W3C. *Web Content Accessibility Guidelines (WCAG) 2.1*. https://www.w3.org/TR/WCAG21/
18. MuscleWiki. *MuscleWiki newsletter: company description*. https://musclewiki.beehiiv.com/